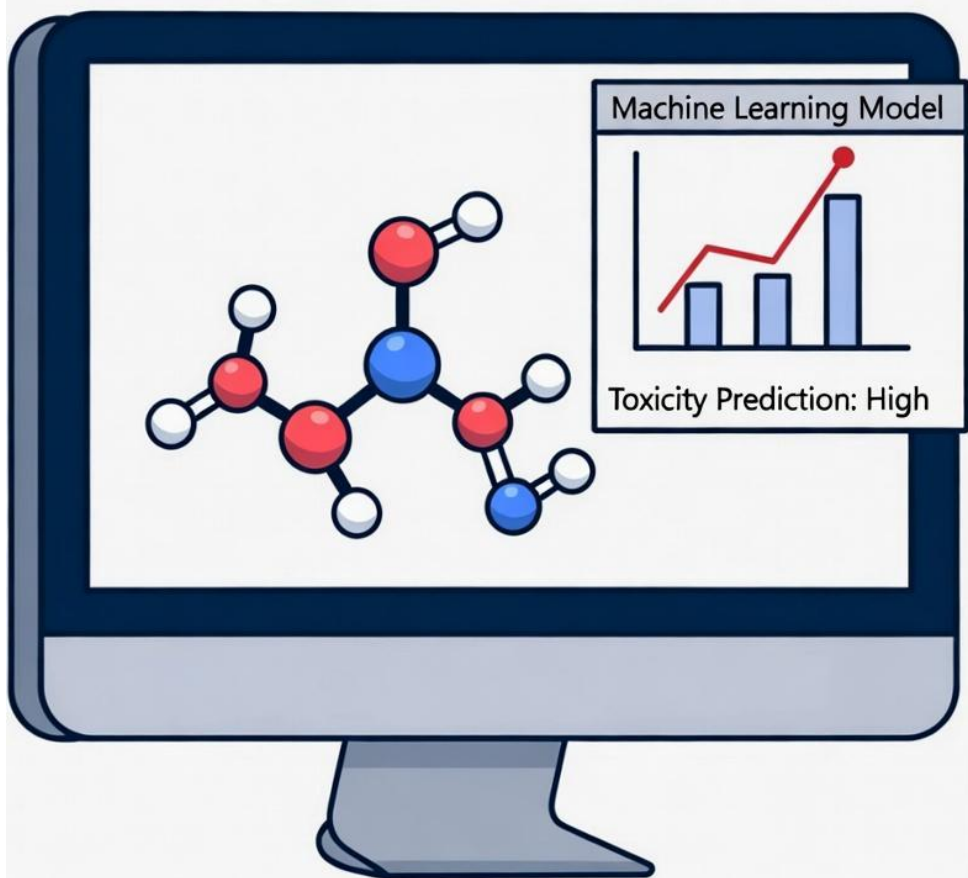




기계학습 (Machine learning)

AI 신약학과

신현길

명령어 (bash, python)

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

새로운 기능 및 개선 사항에 대한 최신 PowerShell을 설치 하세요! https://aka.ms/PSWindows

PS C:\Windows\System32> python
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> python
Traceback (most recent call last):
  File "<python-input-0>", line 1, in <module>
    python
NameError: name 'python' is not defined
>>>
```

코딩을 배운다는 것?

- 예전에는 직접 알고리즘을 작성하는 것
- 최근에는 남이 작성한 알고리즘을 잘 활용하는 것
 - 남이 작성한 코드: 라이브러리 (import OOOO)
- 수업에서 사용하게 될 라이브러리
 - Pandas (csv 파일 처리)
 - RDKit (분자 구조 분석)
 - Scikit-learn (기계학습)
 - Matplotlib (시각화)
 - Optuna (hyperparameter 최적화)

라이브러리 의존성 (library dependency)

Python

1. Pandas (numpy v 1.2)
2. RDKit
3. Scikit-learn
4. Matplotlib
5. Optuna

Anaconda python

base

1. Numpy (v2.1)
2. Jupyter notebook

rdkit

1. Numpy (v1.2)
2. Pandas
3. rdkit
4. Scikit-learn
5. Matplotlib
6. Optuna

라이브러리 설치 주의 사항 (아나콘다 환경)

Windows PowerShell

×

+

▽

Windows PowerShell

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새로운 기능 및 개선 사항에 대한 최신 PowerShell을 설치 하세요! <https://aka.ms/PSWindows>

PS C:\Windows\System32> **pip** install notebook



Anaconda Prompt

×

+

▽

(base) C:\Users\user>jupyter notebook

'jupyter'은(는) 내부 또는 외부 명령, 실행할 수 있는 프로그램, 또는 배치 파일이 아닙니다.

(base) C:\Users\user>

라이브러리 설치 명령어 (command)

pip install OOOOOOO

conda install OOOOOOO

Anaconda python

base

1. Numpy (v2.1)
2. Jupyter notebook

pip install

rdkit

1. Numpy (v1.2)
2. Pandas
3. rdkit
4. Scikit-learn
5. Matplotlib
6. Optuna

How to install RDKit with Conda

Creating a new conda environment with the RDKit installed requires one single command similar to the following::

```
$ conda create -c conda-forge -n my-rdkit-env rdkit
```

Finally, the new environment must be activated so that the corresponding python interpreter becomes available in the same shell:

```
$ conda activate my-rdkit-env
```

If for some reason this does not work, try:

```
$ cd [anaconda folder]/bin  
$ source activate my-rdkit-env
```

Windows users will use a slightly different command:

```
C:\> activate my-rdkit-env
```

Anaconda python

base

1. Numpy (v2.1)
2. Jupyter notebook
3. Scikit-learn
4. matplotlib

my-rdkit-env

1. rdkit

```
Anaconda Prompt - conda cri × + ▾
(base) C:\Users\user>conda create -c conda-forge -n my-rdkit-env rdkit
Do you accept the Terms of Service (ToS) for https://repo.anaconda.com/pkgs/main? [(a)ccept/(r)eject/(v)iew]: a
Do you accept the Terms of Service (ToS) for https://repo.anaconda.com/pkgs/r? [(a)ccept/(r)eject/(v)iew]: a
Do you accept the Terms of Service (ToS) for https://repo.anaconda.com/pkgs/msys2? [(a)ccept/(r)eject/(v)iew]: a
3 channel Terms of Service accepted
Retrieving notices: done
Channels:
- conda-forge
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

==> WARNING: A newer version of conda exists. <==
  current version: 25.11.1
  latest version: 26.1.1

Please update conda by running

  $ conda update -n base -c defaults conda

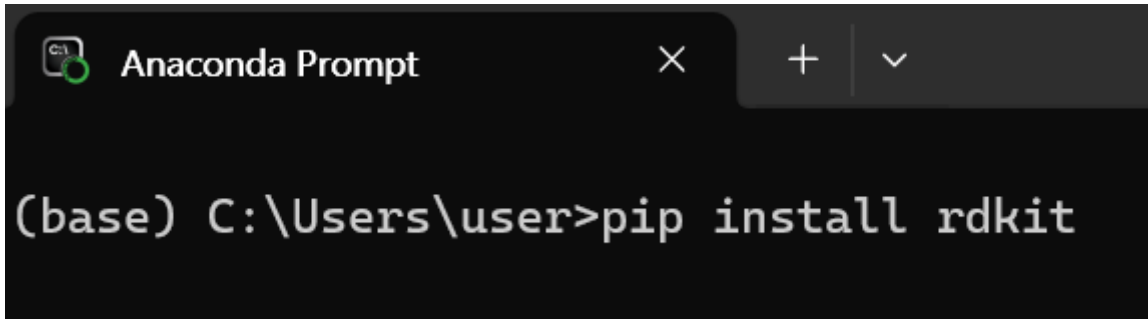
## Package Plan ##

environment location: C:\Users\user\miniconda3\envs\my-rdkit-env

added / updated specs:
```


Anaconda Prompt - conda cri				+	▼	—	□	×
Downloading and Extracting Packages:								
mkl-2025.3.0	95.6 MB	#####2	48%					
python-3.14.3	17.4 MB	#####	100%					
rdkit-2025.09.6	16.8 MB	#####	100%					
pandas-3.0.1	13.4 MB	#####	100%					
icu-78.2	12.6 MB	#####	100%					
librdkit-2025.09.6	11.6 MB	#####	100%					
openssl-3.6.1	8.9 MB	#####	100%					
matplotlib-base-3.10	7.8 MB	#####	100%					
numpy-2.4.2	7.0 MB	#####	76%					
libpq-18.3	4.0 MB	#####	100%					
libglib-2.86.4	3.9 MB	#####	100%					
sqlalchemy-2.0.47	3.8 MB	#####	100%					
tk-8.6.13	3.4 MB	#####	100%					
libhwloc-2.12.2	2.3 MB	#####3	36%					
libboost-1.86.0	2.3 MB	#####9	17%					
reportlab-4.4.10	1.6 MB	#####5	26%					
font-ttf-ubuntu-0.83	1.5 MB		0%					
cairo-1.18.4	1.5 MB		0%					
qhull-2020.2	1.3 MB		0%					
libsqlite-3.51.2	1.2 MB		0%					
libxcb-1.17.0	1.2 MB		0%					
pip-26.0.1	1.1 MB		0%					
pcre2-10.47	973 KB		0%					
libtiff-4.7.1	970 KB		0%					
pillow-12.1.1	951 KB		0%					
libjpeg-turbo-3.1.2	822 KB		0%					
fonttools-4.61.1	815 KB		0%					
libgcc-15.2.0	801 KB		0%					

pip install 사용시 주의사항



```
Anaconda Prompt
(base) C:\Users\user>pip install rdkit
```

Anaconda python

base

1. Rdkit

ds-ml-course

1. Pandas
2. Scikit-learn
3. Matplotlib
4. Optuna

JupyterLab

Install JupyterLab with `pip`:

```
pip install jupyterlab
```

Note: If you install JupyterLab with conda or mamba, we recommend using [the conda-forge channel](#).

Once installed, launch JupyterLab with:

```
jupyter lab
```

Jupyter Notebook

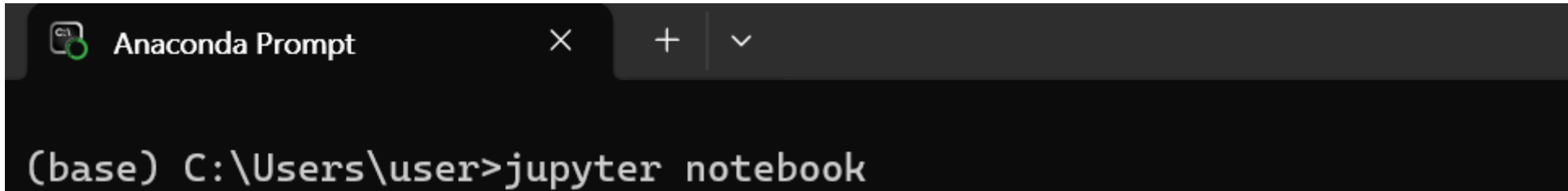
Install the classic Jupyter Notebook with:

```
pip install notebook
```

To run the notebook:

```
jupyter notebook
```

notebook 실행 명령어



```
(base) C:\Users\user>python -m notebook
[I 2026-03-04 05:04:10.265 ServerApp] Extension package jupyter_server_terminals took 3.2252s to import
[I 2026-03-04 05:04:15.314 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2026-03-04 05:04:15.320 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2026-03-04 05:04:15.332 ServerApp] jupyterlab | extension was successfully linked.
[I 2026-03-04 05:04:15.340 ServerApp] notebook | extension was successfully linked.
[I 2026-03-04 05:04:15.351 ServerApp] Writing Jupyter server cookie secret to C:\Users\user\AppData\Roaming\jupyter\runtime\jupyter_cookie_secret
[I 2026-03-04 05:04:15.763 ServerApp] notebook_shim | extension was successfully linked.
[I 2026-03-04 05:04:15.868 ServerApp] notebook_shim | extension was successfully loaded.
[I 2026-03-04 05:04:15.871 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2026-03-04 05:04:15.872 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2026-03-04 05:04:15.876 LabApp] JupyterLab extension loaded from C:\Users\user\miniconda3\Lib\site-packages\jupyterlab
[I 2026-03-04 05:04:15.876 LabApp] JupyterLab application directory is C:\Users\user\miniconda3\share\jupyter\lab
[I 2026-03-04 05:04:15.877 LabApp] Extension Manager is 'pypi'.
[I 2026-03-04 05:04:16.196 ServerApp] jupyterlab | extension was successfully loaded.
[I 2026-03-04 05:04:16.201 ServerApp] notebook | extension was successfully loaded.
[I 2026-03-04 05:04:16.203 ServerApp] Serving notebooks from local directory: C:\Users\user
[I 2026-03-04 05:04:16.203 ServerApp] Jupyter Server 2.17.0 is running at:
[I 2026-03-04 05:04:16.203 ServerApp] http://localhost:8888/tree?token=4d4a717f93ad5ff775894af9c4700e546082d90a94c20276
[I 2026-03-04 05:04:16.203 ServerApp] http://127.0.0.1:8888/tree?token=4d4a717f93ad5ff775894af9c4700e546082d90a94c20276
[I 2026-03-04 05:04:16.203 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2026-03-04 05:04:16.247 ServerApp]
```

To access the server, open this file in a browser:

file:///C:/Users/user/AppData/Roaming/jupyter/runtime/jpserver-9668-open.html

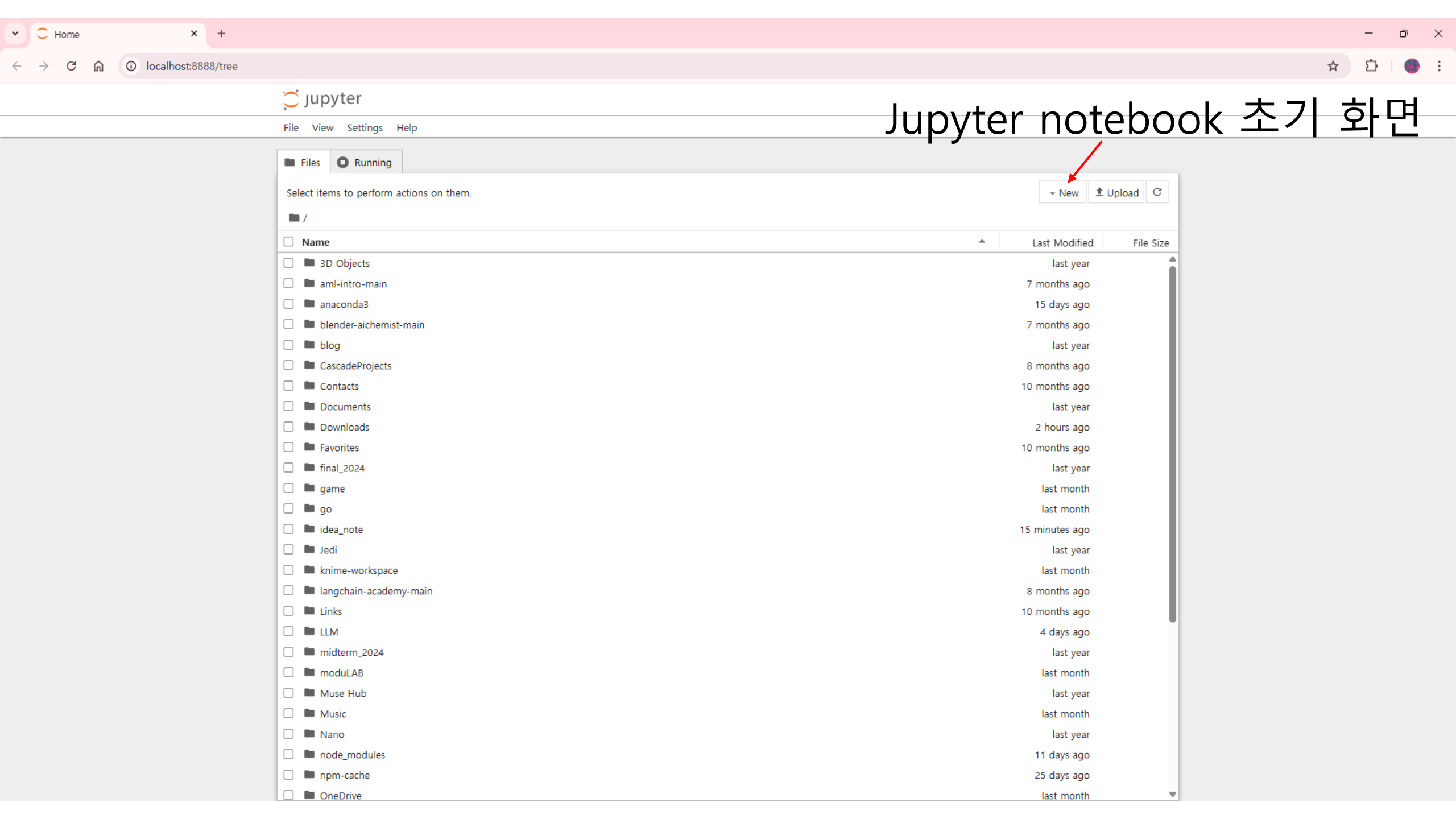
Or copy and paste one of these URLs:

http://localhost:8888/tree?token=4d4a717f93ad5ff775894af9c4700e546082d90a94c20276

http://127.0.0.1:8888/tree?token=4d4a717f93ad5ff775894af9c4700e546082d90a94c20276

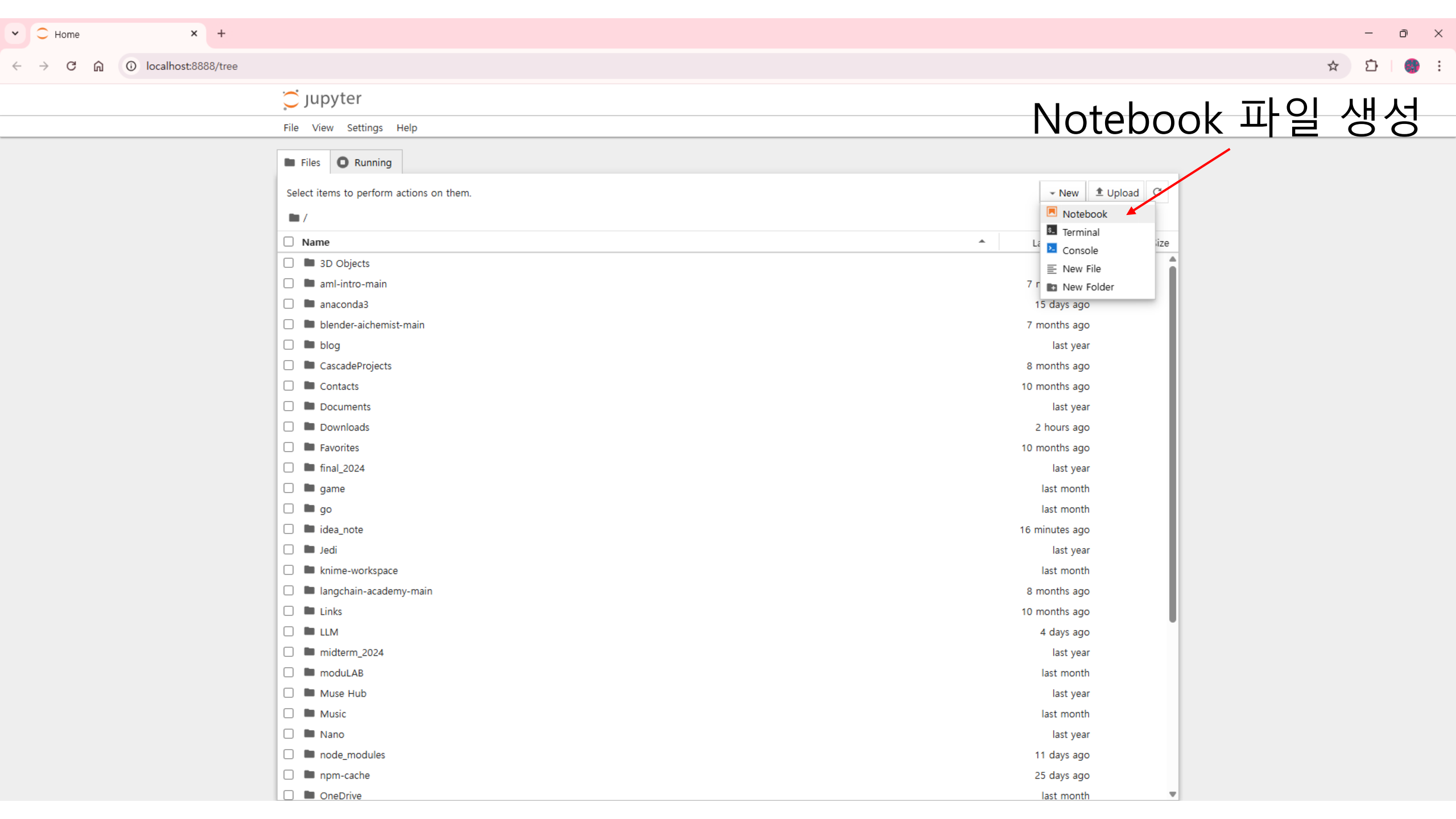
```
[I 2026-03-04 05:04:19.427 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server-nodejs, javascript-typescript-langserver, jed
i-language-server, julia-language-server, pyrefly, pyright, python-language-server, python-lsp-server, r-languageserver, sql-language-server, texlab, typescript-language-server, u
nified-language-server, vscode-css-languageserver-bin, vscode-html-languageserver-bin, vscode-json-languageserver-bin, yaml-language-server
```

Jupyter 실행하면 보이는 창.
창을 닫으면 안됩니다.
Jupyter가 실제로 구동되는
창이에요.

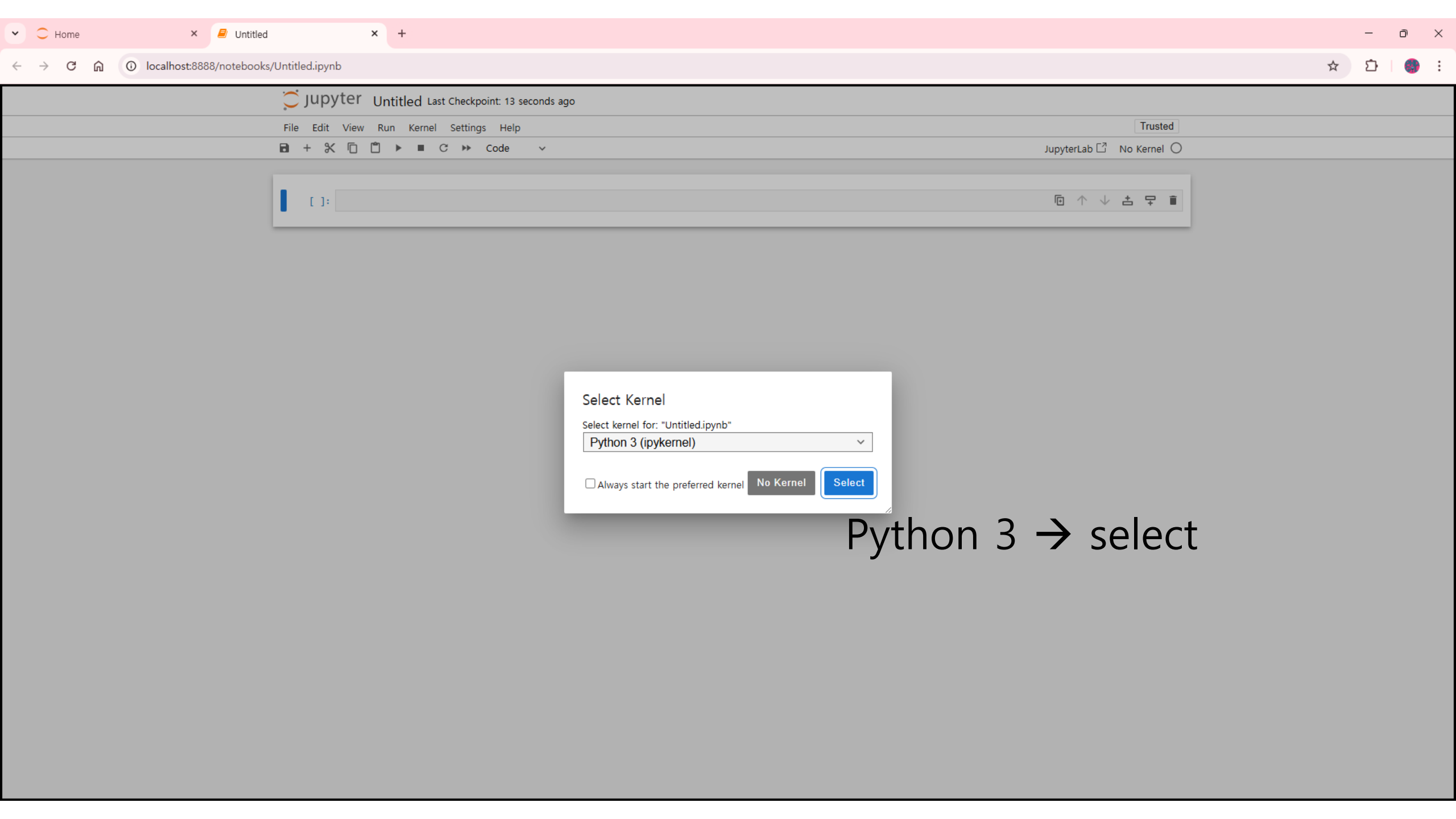


Jupyter notebook 초기 화면

Select items to perform actions on them.		
/		
☐ Name	Last Modified	File Size
☐ 3D Objects	last year	
☐ aml-intro-main	7 months ago	
☐ anaconda3	15 days ago	
☐ blender-aichemist-main	7 months ago	
☐ blog	last year	
☐ CascadeProjects	8 months ago	
☐ Contacts	10 months ago	
☐ Documents	last year	
☐ Downloads	2 hours ago	
☐ Favorites	10 months ago	
☐ final_2024	last year	
☐ game	last month	
☐ go	last month	
☐ idea_note	15 minutes ago	
☐ Jedi	last year	
☐ knime-workspace	last month	
☐ langchain-academy-main	8 months ago	
☐ Links	10 months ago	
☐ LLM	4 days ago	
☐ midterm_2024	last year	
☐ moduLAB	last month	
☐ Muse Hub	last year	
☐ Music	last month	
☐ Nano	last year	
☐ node_modules	11 days ago	
☐ npm-cache	25 days ago	
☐ OneDrive	last month	



Notebook 파일 생성



Select Kernel

Select kernel for: "Untitled.ipynb"

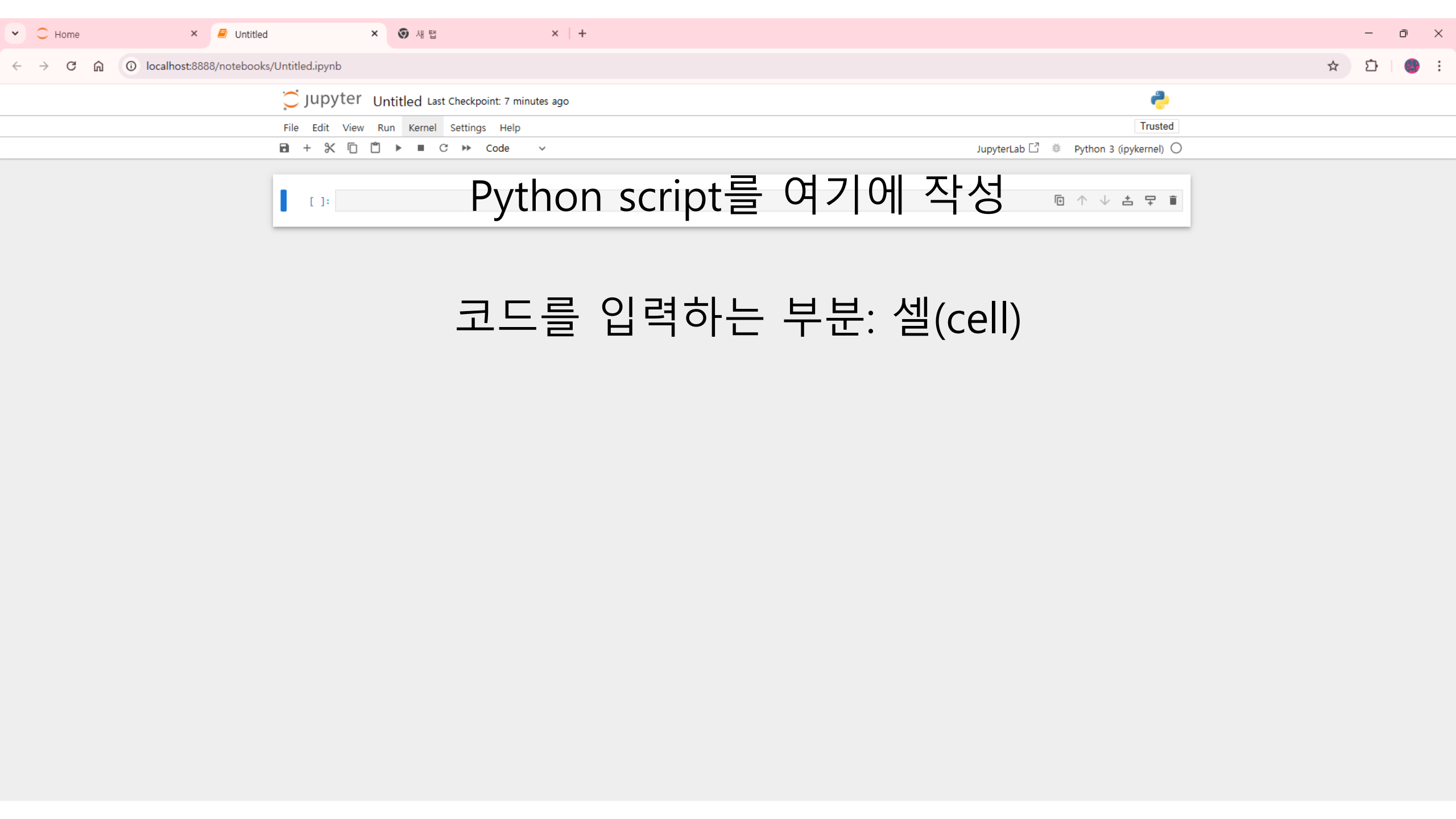
Python 3 (ipykernel)

☐ Always start the preferred kernel

No Kernel

Select

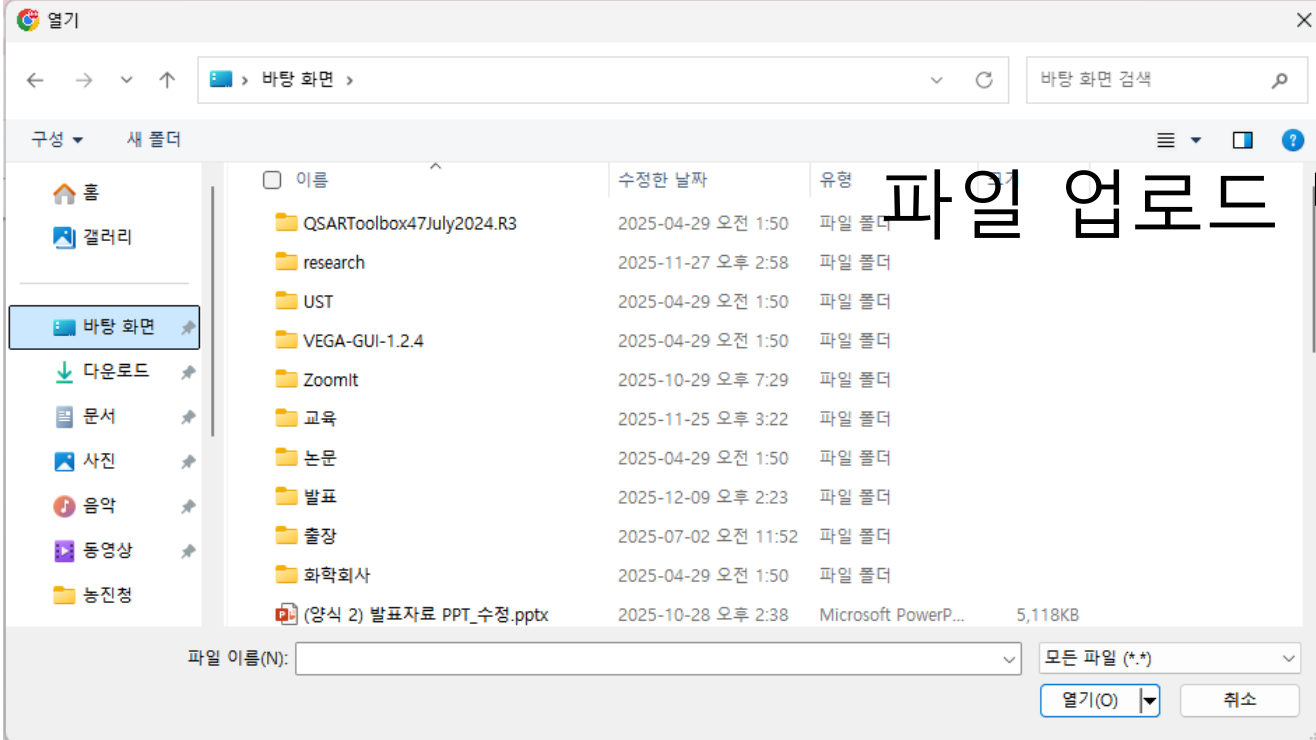
Python 3 → select



Python script를 여기에 작성

코드를 입력하는 부분: 셀(cell)

파일 업로드 방법 (ipynb 파일 및 데이터 파일)



	Last Modified	File Size
	6 days ago	22.4 KB
	16 days ago	29.2 KB
	last year	2.8 KB
	last year	67.1 KB
	8 months ago	8.1 KB
	2 months ago	10.8 KB
	2 months ago	8.8 KB
	2 months ago	617 B
	last year	49.5 KB
	last year	23.7 KB
	last year	2.2 KB
	8 months ago	1.6 KB
	last year	52.5 KB
	last year	5.2 KB
	last year	25.7 KB
	last year	22.7 KB
	4 minutes ago	9 KB
	2 months ago	573.8 KB
	last month	1.4 MB
	last month	4.7 KB
ollama_vision_model.ipynb	3 minutes ago	23.6 KB
open_interpreter.ipynb	8 months ago	46.5 KB
youtube_testing.ipynb	2 months ago	2.5 KB
genAI.env	11 months ago	547 B
NTP_screenshot_1.png	2 months ago	282.9 KB
ollama_gpt_oss.png	last month	12 KB
safety_info.json	last month	133.1 KB

코드 실행 버튼 선택된 셀을 실행합니다.
혹은 셀을 선택한 후 shift+enter

[12]: print('Hello World!')

Hello World!

[5]: text = 'Hello World!'

따옴표(")가 없으면 **상자의 이름**으로 인식.
'따옴표'가 있으면 **상자의 내용물**로 인식.

[6]: text

*상자라고 표현했지만, 프로그래밍에서 공식적으로는 '변수'라고 부릅니다.

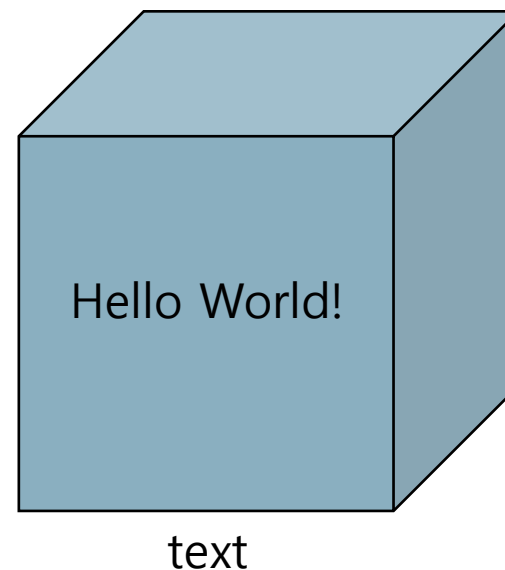
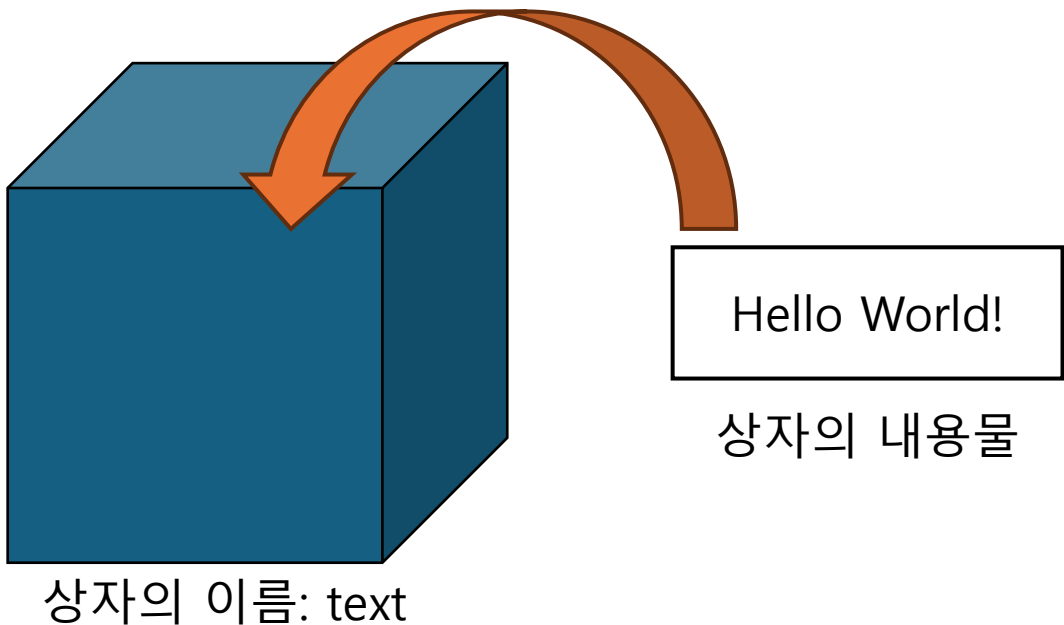
[6]: 'Hello World!'

[7]: print(text)

Hello World!

= (등호의 의미: 상자에 집어넣는다)

Print함수에 상자 이름을 입력해주면
상자의 내용물을 확인할 수 있음.



Pandas data frame

- 데이터 파일 읽기 `pd.read_csv()`, `pd.read_excel()`
- 데이터 분석 목적
 - 기계학습 모델 입력 값이 될 컬럼 이름? (smiles 코드가 있는 컬럼)
 - 기계학습 모델 출력 값이 될 컬럼 이름? (모델이 예측해야 하는 값)
 - 회귀 (regression) / 분류 (classification) 모델? (연속적인 값 / 카테고리)
 - Invitro 탭과 invivo 탭의 차이? (실험 값의 의미, 데이터 양)